

ME 300 Homework 4: Ideal Gases, Non-Ideal Gases, and Phase Change

Answer the following questions neatly on a separate piece of paper or on a tablet. Written answers can be typed. Show all your work, list all your assumptions, and explain your reasoning – partial credit will only be given if you show and explain each of your steps. Submit your homework on Canvas.

- (15 points) Throughout this class we're going to analyze the power and steam production on campus. The East Campus and West Campus Steam Plants are what are called "cogeneration plants" or "combined heat and power plants." Learn more here:
 - Campus power: <https://www.opp.psu.edu/unit/steam-services>
 - Combined heat and power: <https://www.epa.gov/chp/what-chp>

The engine used to make the power is a gas turbine engine – learn about how a gas turbine engine works here: <https://www.youtube.com/watch?v=jrWHi77oAWI>. The engine in this video is very similar to the one we have on campus. In this problem, we'll be analyzing the air going through the compressor.

It's very difficult to directly measure the density of a fluid. Instead, we tend to measure other properties like temperature and pressure and then use an equation of state to calculate density. Download the spreadsheet of data from Canvas that has temperature and pressure at the entrance to the engine (T1 and P1) and after the compressor of the engine (T2 and P2). The working fluid is air, where $R=287 \text{ J/kg}\cdot\text{K}$. Discuss the assumptions you need to calculate the density and then calculate density at both parts of the engine at each hour in the spreadsheet. Hint: be careful about units! Then plot the densities together on the same plot as a function of time. Discuss the trends in the plot – how are things changing with time or day and why do you think it's happening?

- (15 points) Fluids tend to become non-ideal at very high pressures. Let's explore this by using the NIST Webbook and looking at properties of CO_2 as a function of pressure. Fill out the table below for CO_2 where you will calculate the volume using the ideal gas law (the molecular weight of CO_2 is 44.01 kg/kmol) and then by looking it up on the NIST webbook. Also use the value of volume from the NIST webbook to calculate the compressibility, Z , for each condition. Finally, identify the phase of CO_2 from the NIST webbook. After you've filled out the table, explain what's happening – why is CO_2 no longer an idea gas as the pressure increases?

T [K]	P [MPa]	v (ideal) [m ³ /kg]	v (NIST) [m ³ /kg]	Z	Phase (NIST)
300	0.2				
300	4.1				
300	8.3				
500	8.3				

- (20 points) Given the following property data for H_2O , designate the phase (or line or point) in T-v or P-v space (i.e., compressed liquid, liquid-vapor mixture, superheated vapor, saturated liquid, etc.) and find the value(s) of the requested property or properties. Use the steam tables to solve this problem – you can find these in the resources module on Canvas. Indicate the table used, e.g., Table D.3K, etc.

NOTE: This is a very important problem! It will help you develop skill in navigating steam tables and will help you develop conceptual understanding of P-v-T space for a 2-phase substance. So, please spend as much time as necessary to work this problem.

- a. $T = 440 \text{ K}$, $P = 0.733679 \text{ MPa}$, $v = 0.26090 \text{ m}^3/\text{kg}$: Phase = ? $h = ?$ Table = ?
- b. $T = 440 \text{ K}$, $P = 5.00 \text{ MPa}$: Phase = ? $u = ?$ Table = ?
- c. $T = 440 \text{ K}$, $P = 0.100 \text{ MPa}$: Phase = ? $v = ?$ Table = ?
- d. $T = 440 \text{ K}$, $P = 0.733679 \text{ MPa}$, $v = 0.200 \text{ m}^3/\text{kg}$: Phase = ? $h = ?$ Table = ?
- e. $T = 647.096 \text{ K}$, $P = 22.064 \text{ MPa}$: Phase = ? $v = ?$ Table = ?
- f. $T = 700 \text{ K}$, $P = 32 \text{ MPa}$: Phase = ? $\rho = ?$ Table = ?
- g. $T = 605 \text{ K}$, $P = 6.0 \text{ MPa}$: Phase = ? $u = ?$ Table = ?

4. (20 points) 2 kg of water undergoes the cycle in the chart below – fill in the missing state points and calculate change in internal energy (in units of kJ) for each step of the cycle. Then plot the cycle on a T-v diagram, making sure the vapor dome is in the plot.

State	P [MPa]	T [K]	v [m ³ /kg]	ΔU [kJ]
1	5		0.0012864	1-2:
2	10	560		2-3:
3	10		0.020091	3-4:
4	5		0.039446	4-1:

5. (15 points) A rigid vessel contains water; at State 1, the water is at the critical point. Heat is slowly removed from fluid. Answer the following questions about this process:
- a. What is the density of the water at State 1?
 - b. The temperature at State 2 is 500 K. What is the internal energy of the fluid at that state?
 - c. Draw this process on a T-v diagram.
6. (15 points) Water in a piston-cylinder system undergoes an isobaric heat addition process. It starts as a saturated liquid at 438.1 K (state 1) and then heat is added. Make sure to indicate the table you used for each of the questions below.
- h. What is the enthalpy at state 1?
 - i. What is the enthalpy when the water reaches a saturated vapor state?
 - j. What is the enthalpy when the water reaches 700 K? What is the phase?